

Calculations				
	Sheep	Pig	Cow	Goat
Meat Diet Portion	1.12%	3.61%	3.43%	0.94%
Meat MCal Target	10,180.40	32,899.46	31,338.35	8,619.29
Potential Dairy Value attached to each juvenile	132.54	0.00	1015.20	126.90
Value per juvenile	48.00	160.00	640.00	48.00
Adjusted Dairy Value attached to juvenile based on Dairy Utilization	66.27	0.00	507.60	63.45
Culled Mothers Value	14.77	4.41	105.26	14.77
Culled Adult Males	0.74	0.22	5.26	0.74
Final Value per juvenile	63.51	164.63	750.53	63.51
Value MCals	63.51	164.63	750.53	63.51
Offspring required Year ⁻¹	160.30	199.84	41.76	135.72
MCals from meat	10180.40	32899.46	31338.35	8619.29
Max Utilized Mcals allowed from Dairy	10623.20	0.00	21194.92	8611.46
MCals actually Utilized from Dairy	10623.20	0.00	21194.92	8611.46
Percent of diet from their meat	1.12%	3.61%	3.43%	0.94%
Percent of Diet from their Dairy	1.16%	0.00%	2.32%	0.94%
Kg Fleece	1,212.77	0.00	0.00	0.00
1 Year+ (age) skins	59.86	7.90	8.20	50.73
Yearly Offspring skins	292.23	236.03	61.87	247.51
Birthing Events per year	1.00	1.86	1.00	1.00
Offspring Mother ⁻¹ Year ⁻¹	0.56	5.48	0.68	0.56
Mothers Step 1	286.25	36.50	61.40	242.36
Males Step 1	80.15	99.92	20.88	67.86
Females Step 1	80.15	99.92	20.88	67.86
Replacement Mothers per year	57.55	7.60	7.86	48.77
Breeding Males Required	15.03	1.92	3.22	12.72
Additional Mothers	235.59	6.61	29.57	199.63
Males Step 2	65.97	18.09	10.06	55.90
Females 2	65.97	18.09	10.06	55.90
Total Mothers	343.80	44.10	69.27	291.13
Total Males	146.12	118.01	30.93	123.76
Total Females	14.18	81.83	10.82	11.96
Total Offspring	292.23	236.03	61.87	247.51
Total Animals	519.13	245.86	114.25	439.57
Units per Hectare	1.00			
Total Adults per Hectare	2.22	0.34	0.54	2.22
Total Offspring per Hectare	1.09	1.49	0.33	1.09
	0.77%		2.17%	0.65%
Total Adults	594	53	102	503
Total Offspring	292	236	62	248
Days for Maintenance	5.00	220.00	0.00	5.00
Dry Mothers	51.57	0.00	7.40	43.62
Other	4,675.76	0.00	0.00	3,960.23
MCals Dry Mothers	60,232.69	0.00	15,670.28	50,942.62
total Maintenance MCals	64,908.45	0.00	15,670.28	54,902.85

MCals Required				
	Sheep	Pig	Cow	Goat
Gestating	174,026	107,758	103,563	147,395
Lactating	332,008	38,307	111,358	281,201
Maintenance	64,908	0	15,670	54,903
Adult Males	17,553	0	6,825	14,861
Offspring	80,218	237,207	226,428	67,943
TOTAL	668,713.7	383,272.7	463,844.3	566,302.6

Animal Diet Breakdown				
	Sheep	Pig	Cow	Goat
Crop	Barley	Barley	Barley	Barley
Grams ⁻¹ Straw/Chaff	1000.0	0.0	4956.0	1000.0
MCal kg ⁻¹				
Grams ⁻¹ Grain	750.0	750.0	1500.0	750.0
MCal kg ⁻¹				
Days	90.0	90.0	90.0	90.0
Number of Animals	292.2	236.0	61.9	247.5
MCals from Straw/Chaff	37347.6	0.0	39184.3	31632.3
MCals from Grain	59177.5	47795.5	25055.6	50121.6
MCals from other feed	96525.1	47795.5	64239.8	81754.0

Diet Calculations		
Plants Mcal	780,820.414	
Meat Mcal	91,250	
Dairy	40,429.586215676	
	Perfect	
Plants %	85.57%	
Meat %	10.00%	
Dairy %	4.43%	
check	Perfect	
Percent		
Domestic Meat	9.10%	
Dairy	4.43%	
Wild Meat	0.90%	
Crops	77.01%	
Wild Plants	8.56%	
check	Perfect	
new dairy	40429.59	
potential dairy < limit	TRUE	
new limit	90.0%	
the limit greater than that left to satisfy	TRUE	
limit	1	
limit dairy checked	FALSE	
Mcalories		Surplus
Domestic Meat	83,037.500	
Dairy	40,429.586	
Wild Meat	8212.500	
Crops	702,738.372	
Wild Plants	78,082.041	
check	Perfect	
SURPLUS DAIRY	0.000	
This could account for additional % of diet	0.00%	

Herd Modelling

	Sheep	Pig	Cow	Goat				
Mothers Req'd	344	44	69	291				
Males for Slaughter	146	118	31	124				
Females for Slaughter	14	82	11	12				
Offspring available for Consumption	292	236	62	248				
Gestation Req'd	95460	91238	69898	80822				
Lactation Req'd	182119	32435	75159	154192				
Maintenance Req'd	216895	0	20948	183689				
Adult Males	17553	0	6825	14861				
Offspring Required	95999	258450	270971	81308				
Energy Required	608026	382123	443802	514873				
Ewe Fleece Kg Yr-1	774							
Ram Fleece Kg year-1	30							
Juvenile Fleece Kg Yr-1	409							
Kg Fleece Year-1	1213	0	0	0				